

# Helium gas or liquid nitrogen for the first thermal shield of a 4 K cryostat: conjugate heat-transfer budget, coolant operating maps and work ledger

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## Abstract

The first thermal shield of a 4 K cryostat intercepts most of the heat arriving from the 300 K environment. It can be held near 50 K by circulated helium gas or at 77 K by liquid nitrogen. This paper separates that choice into three questions and answers each with a dedicated model. (1) The passive penalty of the warmer shield is computed with a three-dimensional multi-region conjugate heat-transfer model of a representative cryostat section that resolves view-factor radiation and the conduction of a 65-line coaxial harness; the view-factor matrix is shown to require row normalisation in a low-emissivity enclosure, after which the model reproduces the closed-form loads to 4 %. (2) The two coolants are compared in a channelled cold plate at a matched 9.4 W duty over Reynolds numbers 500–10 000 and helium pressures of 1–18 bar, with National Institute of Standards and Technology (NIST) properties, three-grid verification and energy closure below 0.2 %. (3) The routes are priced in room-temperature power with Carnot multipliers and with hardware efficiencies. The warmer shield costs nothing at the first stage, 8.3 against 8.4 W; its penalty is 106 mW at the 4 K stage, 86 % of it harness conduction, so it is a property of the wiring rather than of the cryogen. In the cold plate nitrogen gives five to six times helium’s heat-transfer coefficient and holds the plate within 4 K of its inlet where helium at 1 bar floats 21 K above; the hottest point of the helium plate reaches 77 K only at the top of the laminar range, and at 18 bar helium’s pumping power is 8.5 times below nitrogen’s at matched Reynolds number. The ideal ledger favours nitrogen

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by 21 % for a bare shield and helium by 9 % with multi-layer insulation; with hardware efficiencies and the liquefaction energy of the nitrogen consumed, the equivalent room-temperature power favours helium by 36–51 % at the nominal shield temperatures and by 13–56 % once each route is charged with the plate temperature its own cold plate reaches, the low end at a common laminar flow where the helium plate floats to 67 K. Nitrogen becomes competitive only if the harness conduction to 4 K is cut by about a factor of three. Operating maps, uncertainty bands and the complete case set are provided.

*Keywords:* cryostat, thermal shield, conjugate heat transfer, liquid nitrogen, helium gas, pulse-tube cryocooler, OpenFOAM

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## Nomenclature

### Roman symbols

|                        |                                                              |
|------------------------|--------------------------------------------------------------|
| $A$                    | area [ $\text{m}^2$ ]                                        |
| $c_p$                  | specific heat [ $\text{J kg}^{-1}\text{K}^{-1}$ ]            |
| $D_h$                  | hydraulic diameter [m]                                       |
| $f$                    | Darcy friction factor [-]                                    |
| $F_{ij}$               | view factor from face $i$ to face $j$ [-]                    |
| $h$                    | heat-transfer coefficient [ $\text{W m}^{-2}\text{K}^{-1}$ ] |
| $J$                    | radiosity [ $\text{W m}^{-2}$ ]                              |
| $k$                    | thermal conductivity [ $\text{W m}^{-1}\text{K}^{-1}$ ]      |
| $\dot{m}$              | mass flow rate [ $\text{kg s}^{-1}$ ]                        |
| Mo                     | Mouromtseff number [-]                                       |
| $N$                    | number of MLI layers [-]                                     |
| NTU                    | number of transfer units [-]                                 |
| $Nu$                   | Nusselt number $hD_h/k$ [-]                                  |
| $p$                    | pressure [Pa]                                                |
| $Pr$                   | Prandtl number [-]                                           |
| $\dot{Q}$              | heat load [W]                                                |
| $q$                    | heat flux [ $\text{W m}^{-2}$ ]                              |
| $Re$                   | Reynolds number $\rho U D_h / \mu$ [-]                       |
| $\dot{S}_{\text{gen}}$ | entropy generation rate [ $\text{W K}^{-1}$ ]                |
| $T$                    | temperature [K]                                              |
| $T_0$                  | heat-rejection temperature, 300 K                            |
| $U$                    | mean velocity [ $\text{m s}^{-1}$ ]                          |
| $\dot{W}$              | wall power [W]                                               |
| $\dot{W}_p$            | pumping power [W]                                            |
| $y^+$                  | wall coordinate [-]                                          |

### Greek symbols

|                       |                                                                   |
|-----------------------|-------------------------------------------------------------------|
| $\beta$               | thermal expansion coefficient [ $\text{K}^{-1}$ ]                 |
| $\varepsilon$         | emissivity; HX effectiveness [-]                                  |
| $\eta_i$              | stage efficiency, fraction of Carnot [-]                          |
| $\kappa_{\text{eff}}$ | effective harness conductivity [ $\text{W m}^{-1}\text{K}^{-1}$ ] |
| $\mu$                 | dynamic viscosity [Pa s]                                          |
| $\rho$                | density [ $\text{kg m}^{-3}$ ]                                    |
| $\sigma$              | Stefan–Boltzmann constant                                         |

**Subscripts** 0 ambient; 1 first stage; 2 second (4 K) stage; b bulk; c conduction; r radiation; sat saturation; w wall.

**Acronyms** CFD computational fluid dynamics; CHT conjugate heat transfer; DTT Divertor Tokamak Test facility; GCI grid convergence index; ITER International Thermonuclear Experimental Reactor; LHC Large Hadron Collider; LHD Large Helical Device; MLI multi-layer insulation; NIST National Institute of Standards and Technology; NTU number of transfer units; OVC outer vacuum can; PTFE polytetrafluoroethylene; PTR pulse-tube refrigerator; SIMPLE semi-implicit method for pressure-linked equations; SST shear-stress transport; STL stereolithography (surface file format).

## 1. Introduction

A cryostat that holds a payload at 4 K places at least one thermal shield between the payload and the 300 K vacuum vessel. The shield intercepts the radiation from the vessel and the conduction along supports and wiring, and it does so at a temperature where refrigeration is one to two orders of magnitude cheaper than at 4 K [1, 2, 3]. Two ways of cooling this first shield are in wide use. Large superconducting systems circulate pressurised helium gas through cooling tubes on the shield: the thermal shield of the International Thermonuclear Experimental Reactor (ITER) runs on helium at 80 K and 1.8 MPa [4], the shield of the Divertor Tokamak Test facility (DTT) on the same loop conditions [5], the shields of the Large Helical Device (LHD) on gaseous helium in parallel paths [6], and the shields of the Large Hadron Collider (LHC) on helium at 50–75 K [7, 8]. Magnet and detector cryostats, and the older generation of helium cryostats, use a liquid-nitrogen bath or loop at 77 K [9, 10]; Li et al. made the practical case for it three decades ago, that a boiling cryogen holds the shield temperature constant and carries a stored reserve, whereas a helium-gas shield warms as soon as its refrigerator stops [11]. Compact laboratory cryostats, including the dry dilution refrigerators that host superconducting quantum processors, use the first stage of a helium pulse-tube refrigerator (PTR) at 40–50 K instead [12, 13, 14]. As these laboratory systems grow toward the multi-metre cryostats now being built for quantum platforms [15, 16], they inherit the shield-cooling question that fusion and accelerator cryostats answered with circulated fluids, and the cryogenic plant becomes a dominant part of the system’s energy cost [17, 18, 19].

The published treatments of this choice fall into four strands that do not meet. First, shield design temperatures are optimised with lumped static budgets and refrigeration-power minimisation: Lagrange-multiplier optimisation of nitrogen, neon and helium-gas intercepts on supports and in insulation [11], multi-stage shields for transmission-line cryostats [20], fusion thermal shields [21, 5], and single-shield helium cryostats brought to zero boil-off by load minimisation [22]. These return a stage total or a shield temperature field, from one-dimensional budgets or, for the feeder shields, finite-element models, but not where the wiring load lands or how it couples to the radiation field. Second, wiring loads are measured and modelled per cable [14, 23, 24], again as conduction integrals that assign the harness a single resistance and leave it uncoupled from the radiation field. Third,

surface-to-surface radiation in computational fluid dynamics (CFD) codes has been validated on enclosure benchmarks and applied to fusion vacuum vessels [25], but not coupled to temperature-dependent harness conduction in a cryostat shield, and the closure of agglomerated view-factor matrices in such enclosures has received little attention. Fourth, cryogenic heat exchangers are optimised by entropy-generation minimisation [26, 27, 28] and liquid-nitrogen cooled components are now modelled with combined analytical and CFD frameworks [29], yet helium gas and liquid nitrogen have not been ranked against each other as forced-convection first-shield coolants on a common duty across flow regimes and pressures, with heat-transfer, hydraulic and effectiveness metrics reported together and the single-phase margin of the nitrogen verified. Finally, the cryocooler and liquefier efficiencies that separate an ideal Carnot ledger from a real one are well documented [30, 8, 31], but the two ledgers have not been reconciled for this choice.

The gap is therefore not a missing number but a missing junction. What is absent is a treatment that (i) resolves shield radiation and harness conduction together in three dimensions on one geometry, (ii) compares the two coolants like for like in the cold plate that geometry implies, and (iii) prices both routes with ideal and with hardware efficiencies, so that the conditions under which each route wins can be stated rather than assumed.

This paper therefore asks three questions about a representative 4 K cryostat section with a 65-line coaxial harness (Fig. 1):

**Q1 (passive penalty).** What does raising the first shield from 50 K to 77 K cost in heat load at the first stage and at the 4 K stage, and which mechanism carries the cost?

**Q2 (coolant performance).** If the first-stage duty is removed by a coolant circulated through a channelled cold plate, how do helium gas and liquid nitrogen compare across flow regimes, pressures and bases of comparison, and where is each fluid’s best operating point?

**Q3 (work ledger).** What do the two routes cost in room-temperature power, with ideal Carnot multipliers and with the efficiencies of real hardware?

Q1 is answered with a three-dimensional multi-region conjugate heat-transfer (CHT) model that resolves view-factor radiation and temperature-dependent harness conduction. Its stage-integrated loads are verified against closed-form budgets; its contribution is the spatial distribution of the load and the

split between harness families. Q2 is answered with a grid-converged cold-plate model run for both fluids over  $Re = 500\text{--}10^4$  and helium pressures of 1–18 bar, with properties from the NIST reference equations of state [32]. Q3 combines the stage loads of Q1 with Carnot multipliers and with published cryocooler and liquid-nitrogen plant efficiencies [30, 33, 31]. The dilution unit or any payload below 4 K is outside the scope; the results apply to any 4 K cryostat with a wired first shield.

## 2. Models and methods

### 2.1. Governing equations

Both models solve steady conservation equations with OpenFOAM v2412 [34]. In the enclosure model the inter-stage gaps are vacuum, so each gap region carries only the radiative exchange and each solid region carries conduction with a temperature-dependent conductivity,

$$\nabla \cdot [\kappa_{\text{eff}}(T) \nabla T] = 0 \quad \text{in the harness bundles,} \quad (1)$$

coupled at the bundle surfaces to the radiative flux of the enclosing gap. Radiation is solved as a grey-diffuse radiosity system on the agglomerated boundary faces,

$$J_i = \varepsilon_i \sigma T_i^4 + (1 - \varepsilon_i) \sum_j F_{ij} J_j, \quad q_{r,i} = J_i - \sum_j F_{ij} J_j, \quad (2)$$

where  $F_{ij}$  are geometric view factors computed by ray tracing between agglomerated faces (`viewFactor` model). For two parallel grey surfaces Eq. (2) reduces to the closed form  $q_r = \sigma(T_{\text{hot}}^4 - T_{\text{cold}}^4)/(1/\varepsilon_{\text{hot}} + 1/\varepsilon_{\text{cold}} - 1)$  [35], which is used as the verification anchor.

The cold-plate model solves the steady Reynolds-averaged (or, in the laminar range, the plain) continuity, momentum and energy equations for a single-phase coolant,

$$\nabla \cdot (\rho \mathbf{U}) = 0, \quad (3)$$

$$\nabla \cdot (\rho \mathbf{U} \mathbf{U}) = -\nabla p + \nabla \cdot [(\mu + \mu_t)(\nabla \mathbf{U} + \nabla \mathbf{U}^T)] + \rho \mathbf{g}, \quad (4)$$

$$\nabla \cdot (\rho \mathbf{U} h) = \nabla \cdot \left[ \left( \frac{k}{c_p} + \frac{\mu_t}{Pr_t} \right) \nabla h \right], \quad (5)$$

with  $\mu_t = 0$  for laminar runs and  $\mu_t$  from the  $k\text{--}\omega$  shear-stress transport (SST) model [36] for  $Re \geq 5000$  ( $Pr_t = 0.85$ ). The sweep is run as pure forced

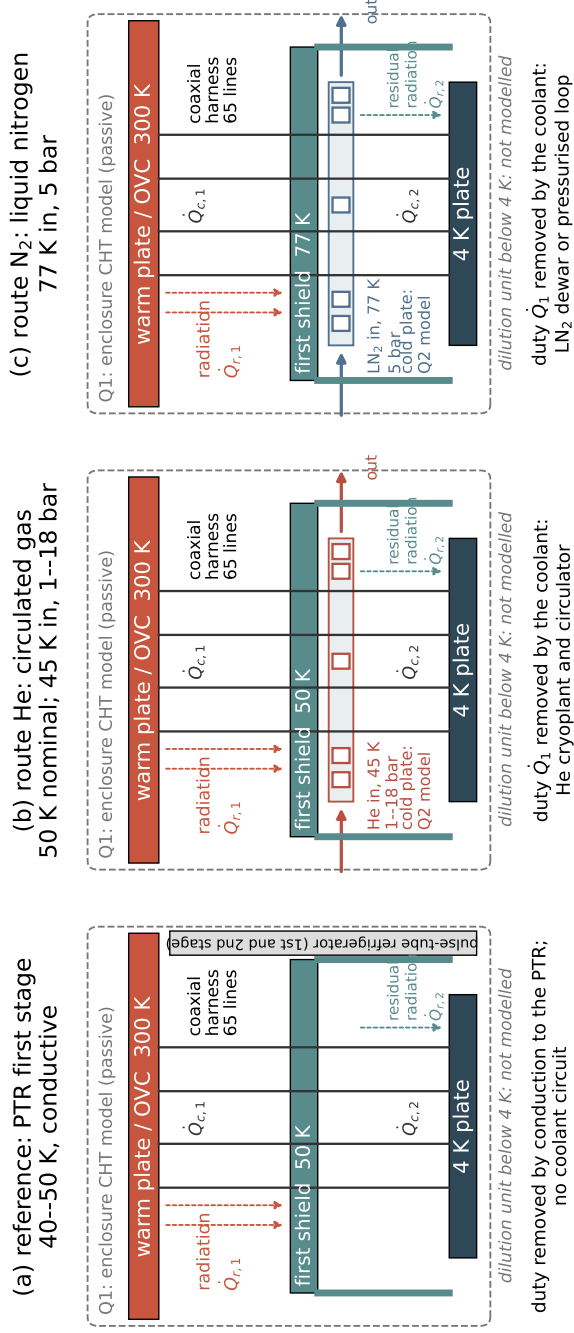


Figure 1: The three first-shield cooling routes and the two models, drawn for the cryostat class studied: a 300 K warm plate and outer vacuum can, a first shield at 50 or 77 K and a 4 K plate, with the dilution unit or payload below 4 K not modelled. (a) Reference architecture with a conductively coupled PTR first stage; (b) circulated helium gas through a channelled cold plate; (c) liquid nitrogen through the same cold plate. The dashed box is the domain of the passive enclosure CHT model (Q1); the shaded channel row is the domain of the active cold-plate model (Q2). The loads  $\dot{Q}_1 = \dot{Q}_{r,1} + \dot{Q}_{c,1}$  and  $\dot{Q}_2 = \dot{Q}_{r,2} + \dot{Q}_{c,2}$  enter the work ledger ( $Q3$ ),  $\dot{W} = \sum_i \dot{Q}_i (T_0 - T_i) / (T_i \eta_i)$ , with  $\eta_i = 1$  (ideal) or the measured fraction of Carnot (real).

convection ( $\mathbf{g} = 0$ ) so that the ranking does not depend on the orientation of the plate; the buoyancy parameter  $\text{Gr}/\text{Re}^2$  is reported for every run, and four runs with gravity acting along the heated wall quantify the mixed-convection sensitivity (Section 4). Helium is treated as an ideal gas with variable density,  $\rho = pM/(R_u T)$ , because its density changes by 5–26 % between inlet and outlet and by up to a factor of 2 between bulk and heated wall; liquid nitrogen is treated with the Boussinesq approximation,  $\rho = \rho_0[1 - \beta(T - T_0)]$ , for which  $\beta\Delta T \leq 0.05$  in every run (largest at  $\text{Re} = 500$ , where the wall sits 8.7 K above the inlet), within the first-order range of the approximation.

## 2.2. Enclosure CHT model (Q1)

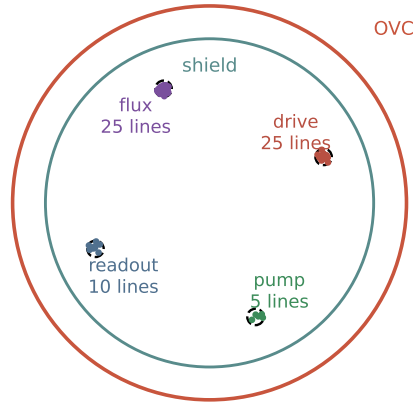
The domain is a  $0.718 \times 0.718 \times 0.335$  m section of a cryostat between the warm plate and the first-stage cold plate (Fig. 3; Table 1). Two vacuum gap regions (outer: outer vacuum can (OVC) to shield; inner: warm plate to cold plate and shield) and four solid cylinders of 20 mm diameter spanning the 0.335 m height are meshed with a  $30 \times 30 \times 14$  background grid refined to level 2 around the cylinders ( $0.2\text{--}0.4 \times 10^6$  cells in total). Each cylinder lumps one functional family of a cabled 65-line control harness [14]: 25 drive, 25 flux, 10 readout and 5 pump lines, all of the UT-085 stainless-steel/PTFE coaxial construction. The family is represented by an effective conductivity

$$\kappa_{\text{eff}}(T) = \frac{n_k}{A_{\text{model}}} \sum_i A_i k_i(T), \quad (6)$$

where  $n_k$  is the number of lines in the family,  $A_i$  the cross-section of cable component  $i$  (outer conductor  $1.585 \text{ mm}^2$  and centre conductor  $0.205 \text{ mm}^2$  of 304 stainless steel, polytetrafluoroethylene (PTFE) dielectric  $2.001 \text{ mm}^2$ ),  $k_i(T)$  the NIST cryogenic conductivity fits, and  $A_{\text{model}}$  the conduction cross-section of the modelled cylinder in the stage span concerned, calibrated from the mesh so that the resolved conduction reproduces the harness integral  $n_k L^{-1} \sum_i A_i \int k_i dT$  (Supplementary Tables S3 and S4 list the areas, lengths, per-bundle targets and the fifth-order polynomial coefficients of  $\kappa_{\text{eff}}$  for every bundle and stage span; fit error below 0.5 %; Fig. 2). The stage-integrated conduction is therefore fixed by the cable data by construction, and the model is credited only with its distribution among bundles and its coupling to the radiation field, not with its total.

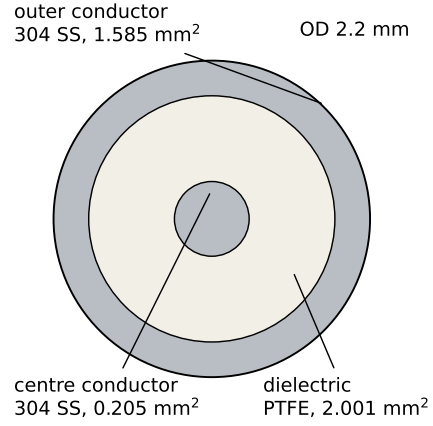
*Verification and uncertainty.* Stage-integrated radiation is compared with the parallel-plate closed form and harness conduction with the analytic integral; both agree within the tolerances reported in Section 3. Because the

(a) 65 lines in four families



each family → one 20 mm cylinder  
at  $r = 0.15$  m (plan view, to scale)

(b) UT-085 cable cross-section



$$(c) \kappa_{eff}(T) = \frac{n_k}{A_{model}} \sum_i A_i k_i(T)$$

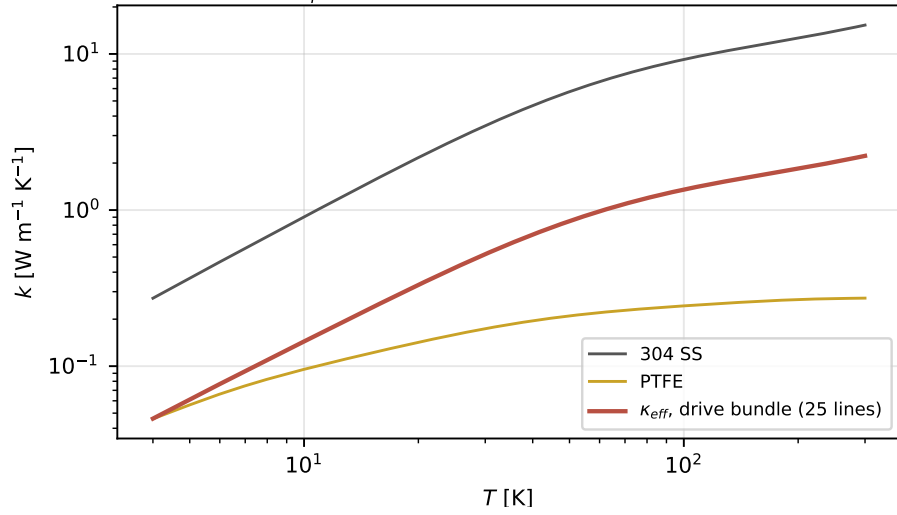


Figure 2: Harness representation: (a) the 65 coaxial lines in four functional families, each lumped into one 20 mm cylinder; (b) the UT-085 cable cross-section with the component areas of Eq. (6); (c) the component conductivities and the resulting effective conductivity of the drive bundle.

Table 1: Enclosure model: geometry, boundary conditions and numerical settings.

|                       |                                                                                                                                                                                                                                   |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Domain                | $0.718 \times 0.718 \times 0.335$ m; OVC and shield as snapped stereolithography (STL) surfaces; four coax cylinders $\varnothing 20$ mm at radius 0.15 m                                                                         |
| Regions               | <b>domain0</b> (OVC–shield gap), <b>domain1</b> (warm plate–cold plate/shield gap), <b>coax.L1..L4</b> (solids)                                                                                                                   |
| Gas in gaps           | vacuum: $p = 10^{-3}$ Pa, $g = 0$ , $\mu = 10^{-11}$ Pa s (no convection, negligible gas conduction; the residual-gas bound is free-molecular helium at $10^{-4}$ Pa, 24 mW or 2.3 % of the harness conduction, Supplementary S1) |
| Temperatures          | fixed: warm plate and OVC at $T_{\text{hot}}$ ; shield and cold plate at $T_{\text{cold}}$ ; cases (300, 50), (300, 77), (50, 4), (77, 4) K                                                                                       |
| Emissivity (baseline) | warm surfaces 0.04 (polished Al); shield and cold plate 0.02 (polished Cu-plated steel); harness 0.02                                                                                                                             |
| Emissivity (variants) | $\pm 50$ % (0.02/0.01 and 0.06/0.03); multi-layer-insulation (MLI) equivalent 0.003 on all first-stage surfaces                                                                                                                   |
| Radiation             | <b>viewFactor</b> , grey diffuse, one band, solved every 10 iterations; agglomeration 250 faces per patch at the coarsest level (verification at 400 and 800)                                                                     |
| Harness contact       | bundles fixed at $T_{\text{hot}}$ at the warm cap and $T_{\text{cold}}$ at the cold cap; no contact resistance; lateral surfaces exchange by radiation only                                                                       |
| Solver                | <b>chtMultiRegionSimpleFoam</b> , semi-implicit method for pressure-linked equations (SIMPLE), residuals $< 10^{-7}$ , 2900 iterations from a uniform initial field                                                               |

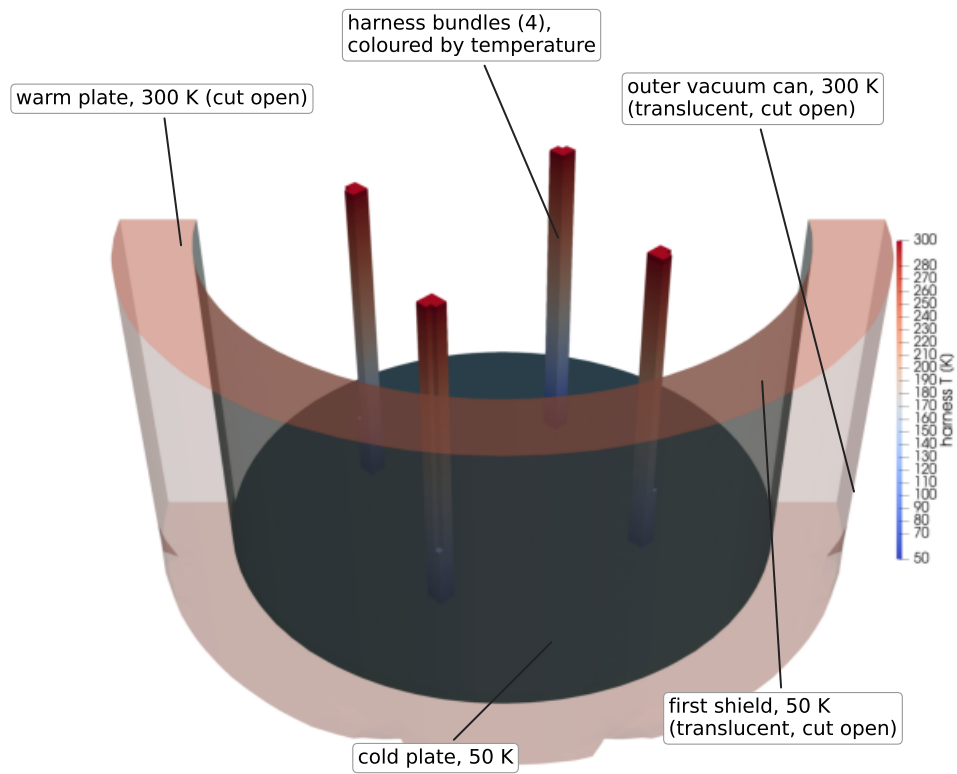


Figure 3: Cut-away of the enclosure CHT domain for the  $300 \rightarrow 50$  K case. The warm plate, outer vacuum can and first shield are cut open on the near side; the four harness bundles, coloured by their computed temperature, span the gap from the warm plate to the cold plate. The vacuum gaps between the surfaces carry radiation only.

view-factor matrix is built on agglomerated faces, the radiative load was re-computed at 250, 400 and 800 coarse faces per patch and its closure  $\sum_j F_{ij}$  monitored (Supplementary Table S5). The emissivity of polished cryogenic surfaces is uncertain by roughly a factor of 1.5 [37]; the  $\pm 50\%$  runs give the uncertainty band carried through the ledger. An MLI-equivalent variant with  $\varepsilon = 0.003$  on the first-stage surfaces represents a 10–20 layer blanket [38, 1].

*What the three-dimensional model adds..* For a parallel-plate enclosure with fixed wall temperatures, the stage-integrated radiative and conductive loads are analytic, and the CHT model must reproduce them; that agreement is the model’s verification, not its result. The model contributes what the closed forms cannot: the distribution of radiative flux over the shield (which sets local shield temperatures once a real, finitely conductive shield is considered), the coupling between the harness and the surrounding radiation field, the first-stage duty that sets the cold-plate load, and the dependence of the 4 K-stage load on the first-stage temperature that closes the loop between the cold-plate and ledger models (Section 5). The partition of conduction among harness families follows from the cable counts and is not claimed as a result. For the cold plate the closed forms are the fully developed duct limits and the Gnielinski correlation; what the model adds there is the developing-flow Nusselt map across Reynolds number and pressure, the departure of a single-wall-heated duct from the correlations for uniformly heated ducts, and the wall-temperature distributions, including the local maximum that sets the saturation margin, none of which a correlation supplies (Section 4).

### 2.3. Cold-plate heat-exchanger model (Q2)

*Basis of comparison..* The two fluids differ in phase, temperature, density by a factor of 800 and thermal conductivity by a factor of 3.1, so no single common basis is neutral. Three bases are therefore reported: matched Reynolds number (same flow regime), matched mass flow, and matched pumping power. All runs remove the same duty of 9.4 W, the conservative first-stage load of Section 3. Helium enters at 45 K, at 1, 5 or 18 bar (the ITER shield loop runs at 18 bar [4]); nitrogen enters as a liquid at 77 K in a 5 bar loop, for which  $T_{\text{sat}} = 94.0$  K (87.9 K at 3 bar). The Boussinesq solution does not depend on the loop pressure, since the 77 K liquid properties change by less than 0.3 % between 3 and 5 bar, so one set of fields serves both pressures; the saturation margin  $T_{\text{sat}} - T_{\text{w,max}}$ , with  $T_{\text{w,max}}$  the hottest face

of the heated-wall patch, is reported for every run to confirm single-phase operation. Table 2 lists the operating quantities and Fig. 4 the geometry.

*Property reference state..* The helium transport properties are the NIST values at 50 K and 1 bar,  $\mu = 6.36 \mu\text{Pa s}$  and  $k = 0.0467 \text{ W m}^{-1} \text{ K}^{-1}$  ( $\text{Pr} = 0.709$ ), held constant in each run with the density from the ideal-gas law; the inlet is 45 K and the wall reaches 106 K at  $Re = 500$ , over which range  $\mu$  and  $k$  both rise by about 65 %. Two sensitivity runs with cubic  $\mu(T)$  and  $k(T)$  fitted to the NIST data over 40–100 K (fit error below 0.05 %) are reported in Section 4.1. The reference state matters: gas properties taken 35 K too cold would understate  $h$  by a factor of about 2.5.

*Metrics..* The heat-transfer coefficient is  $h = \dot{Q}/[A_{\text{heated}}(T_w - T_b)]$  with  $T_w$  the area-averaged heated-wall temperature and  $T_b$  the mean of the inlet and the mass-flux-weighted outlet temperature;  $Nu = hD_h/k$ ; the Darcy friction factor  $f = 2\Delta p D_h/(\rho U^2 L)$ ; pumping power  $\dot{W}_p = \dot{m}\Delta p/\rho$ ;  $\text{NTU} = hA/(\dot{m}c_p)$  and  $\varepsilon = 1 - e^{-\text{NTU}}$  [39]; the Mouromtseff number  $\text{Mo} = k^{0.6}\rho^{0.8}c_p^{0.4}\mu^{-0.4}$  [40]. Laminar results are checked against the fully developed square-duct limits for one heated and three adiabatic walls,  $Nu_\infty \approx 2.71$  [41] (a finite-difference solution of the fully developed problem on the present cross-section gives 2.69; the four-wall value is 3.61) and  $fRe = 56.9$ , and turbulent results are compared with the Gnielinski correlation [42], which is for uniformly heated ducts and serves as a reference rather than a benchmark. Wall temperatures are taken on the heated-wall patch itself: its area mean, its hottest and coldest faces, and the face-value profile along the centre row of the first channel.

*Grid convergence..* Three systematically refined meshes (refinement ratio 1.5) were run for both fluids at  $Re = 2300$  and the discretisation uncertainty expressed as a GCI [43] (Section 4). For the turbulent runs the first-cell  $y^+$  on the graded mesh averages 2.5–5.6 with a maximum of 9.5, within the range handled by the low-Reynolds wall treatment of the SST model.

#### 2.4. Work ledger ( $Q3$ )

A load  $\dot{Q}_i$  absorbed at  $T_i$  and rejected at  $T_0 = 300 \text{ K}$  costs

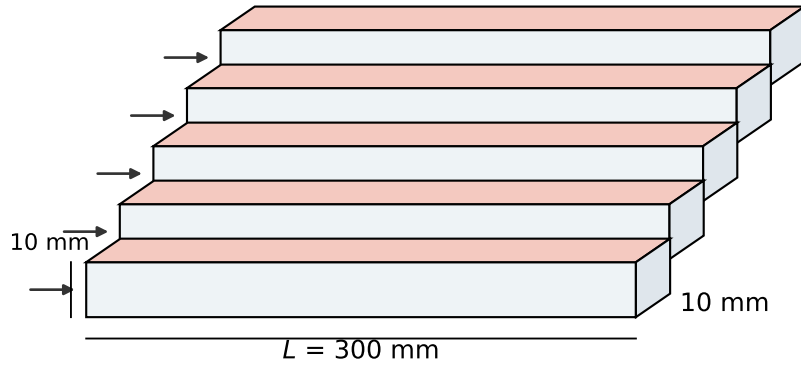
$$\dot{W}_i = \frac{\dot{Q}_i}{\eta_i} \frac{T_0 - T_i}{T_i}, \quad (7)$$

Table 2: Cold-plate model: geometry, coolant states, properties and numerics. Properties from the NIST reference equations of state via CoolProp [32], evaluated at the inlet state and held constant except for the helium density (ideal gas) and the nitrogen buoyancy term.

| <b>Shared geometry and duty</b>                 |                                                                                                                                            |                                                              |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Channels                                        | five parallel square ducts, one wall heated                                                                                                |                                                              |
| Dimensions                                      | $D_h = 10 \text{ mm}$ , $L = 0.30 \text{ m}$ ; $A_{\text{heated}} = 0.015 \text{ m}^2$ ,<br>$A_{\text{cs}} = 5 \times 10^{-4} \text{ m}^2$ |                                                              |
| Duty                                            | 9.4 W, uniform flux $q_w = 627 \text{ W m}^{-2}$ on the heated wall                                                                        |                                                              |
| Mesh                                            | $180 \times 18 \times 18$ cells per duct, wall-graded (ratio 6); $2.92 \times 10^5$ cells; GCI on 86k / 292k / 984k                        |                                                              |
| Gravity                                         | off (forced convection); on for three sensitivity runs                                                                                     |                                                              |
|                                                 | <b>Helium gas</b>                                                                                                                          | <b>Liquid nitrogen</b>                                       |
| <i>Coolant state</i>                            |                                                                                                                                            |                                                              |
| Inlet temperature                               | 45 K                                                                                                                                       | 77 K                                                         |
| Loop pressure                                   | 1, 5 or 18 bar                                                                                                                             | 5 bar ( $T_{\text{sat}} = 94.0 \text{ K}$ ; 87.9 K at 3 bar) |
| Phase                                           | ideal gas, variable density                                                                                                                | single-phase liquid (margin reported)                        |
| <i>Properties at the inlet state</i>            |                                                                                                                                            |                                                              |
| Density $\rho$ [ $\text{kg m}^{-3}$ ]           | 0.961 / 4.80 / 17.3 at 1 / 5 / 18 bar                                                                                                      | 808                                                          |
| $c_p$ [ $\text{J kg}^{-1} \text{K}^{-1}$ ]      | 5201                                                                                                                                       | 2038                                                         |
| Viscosity $\mu$ [ $\mu\text{Pa s}$ ]            | 6.36                                                                                                                                       | 163.5                                                        |
| $k$ [ $\text{W m}^{-1} \text{K}^{-1}$ ]         | 0.0467                                                                                                                                     | 0.1457                                                       |
| Prandtl number $Pr$                             | 0.71                                                                                                                                       | 2.29                                                         |
| $\beta$ [ $\text{K}^{-1}$ ]                     | $1/T$ (ideal gas)                                                                                                                          | $5.6 \times 10^{-3}$                                         |
| <i>Flow range</i>                               |                                                                                                                                            |                                                              |
| Reynolds numbers                                | 500, 1000, 1500, 2300 (laminar); 5000, 10 000 ( $k$ - $\omega$ SST)                                                                        |                                                              |
| $\dot{m}$ at $Re = 2300$ [ $\text{kg s}^{-1}$ ] | $7.31 \times 10^{-4}$                                                                                                                      | $1.88 \times 10^{-2}$                                        |
| $U$ at $Re = 2300$ [ $\text{m s}^{-1}$ ]        | 1.52 (1 bar); 0.084 (18 bar)                                                                                                               | 0.047                                                        |
| <i>Numerics</i>                                 |                                                                                                                                            |                                                              |
| Solver                                          | buoyantSimpleFoam (ideal gas)                                                                                                              | buoyantBoussinesqSimpleFoam                                  |
| Convergence                                     | residuals $< 10^{-6}$                                                                                                                      | residuals $< 10^{-7}$                                        |
| Energy closure                                  | per run; $< 0.2 \%$ for every forced-convection case                                                                                       |                                                              |

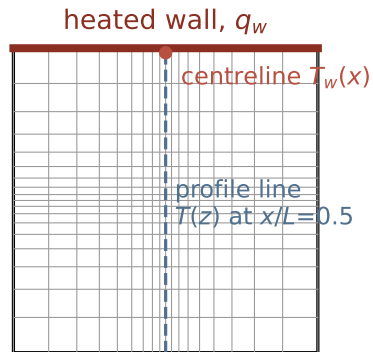
(a) cold-plate heat-exchanger domain

heated wall (top of every duct): uniform  $q_w = \dot{Q}/A_{heated} = 627 \text{ W m}^{-2}$



inlet:  $T_{in}$ ,  $\dot{m}$  (He: mass-flow BC; LN<sub>2</sub>: velocity BC); outlet: fixed  $p$   
 other walls adiabatic and no slip; five square ducts,  $D_h = 10 \text{ mm}$ , pitch 18 mm  
 heated area  $0.015 \text{ m}^2$  (top walls), flow area  $5 \times 10^{-4} \text{ m}^2$

(b) duct cross-section: mesh and sampling lines



18×18 cells per duct  
 two-sided grading, ratio 6  
 first cell 0.19 mm  
 180 cells along  $L$   
 $2.9 \times 10^5$  cells in total

Figure 4: Cold-plate heat-exchanger model: (a) the five heated square ducts with boundary conditions and dimensions; (b) the graded duct cross-section with the two sampling lines used for the wall-temperature profiles.

198 where  $\eta_i$  is the stage efficiency as a fraction of Carnot. The ideal ledger sets  
 199  $\eta_i = 1$ : the multiplier is 5.0 at 50 K, 2.9 at 77 K and 74 at 4 K. The real ledger  
 200 uses published hardware. A commercial two-stage PTR (Cryomech PT415)  
 201 delivers 1.5 W at 4.2 K together with 40 W at 45 K for 10.7 kW of input [33],  
 202 so the two stages together return 3.1 % of their combined Carnot work. A  
 203 single machine cannot separate the two stage efficiencies, so the ledger takes  
 204 them from Radebaugh’s survey [30]:  $\eta_2 = 0.008$ – $0.015$  at 4 K and  $\eta_1 = 0.10$ –  
 205  $0.15$  at 45–80 K (Supplementary S3). These ranges are consistent with the  
 206 PT415: at their midpoints the two stages would need 11.4 kW, within 7 % of  
 207 the rated input. Liquid nitrogen carries  $199 \text{ kJ kg}^{-1}$  of latent heat at 1 bar;  
 208 its ideal production work is  $0.213 \text{ kWh kg}^{-1}$  [2] and real air-separation plants  
 209 need  $0.5$ – $0.7 \text{ kWh kg}^{-1}$  [31], an exergy efficiency  $\eta_{\text{N}_2} \approx 0.3$ – $0.4$ . The 4 K stage  
 210 of the nitrogen route is still served by a cryocooler, so its  $\eta_2$  is the same in  
 211 both routes. The ranges above are propagated as bands.

### 212 **3. Results Q1: the passive penalty of a 77 K shield**

#### 213 *3.1. Radiation verification and the view-factor closure*

214 The first-stage radiative load computed with the view-factor matrix as  
 215 generated does not converge under agglomeration refinement: 9.46, 8.41, 6.49  
 216 and 3.26 W at 60, 250, 400 and 800 coarse faces per patch, against 7.07 W  
 217 from the parallel-plate closed form (Fig. 5). The cause is closure. In a closed  
 218 enclosure every row of the view-factor matrix must sum to one; the generated  
 219 rows sum to between 0.50 and 1.13 (mean 0.97–1.01), and finer agglomeration  
 220 worsens the scatter. With emissivities of 0.02–0.04 the radiosity is reflected  
 221 of order  $1/\varepsilon$  times before absorption, so a per-reflection leak of a few per cent  
 222 compounds into a load error of tens of per cent. Normalising each row to unity  
 223 removes the artefact: at the production level of 250 faces the  $300 \rightarrow 50 \text{ K}$  load  
 224 becomes 7.36 W (4.1 % above the closed form), the  $300 \rightarrow 77 \text{ K}$  load 7.34 W  
 225 (4.2 %), and the two 4 K-stage loads 3.16 and 17.8 mW (0.5 % above 3.15  
 226 and 17.7 mW). The normalised load converges under refinement, 7.36, 7.32  
 227 and 7.27 W at 250, 400 and 800 faces, so the remaining 3–4 % combines the  
 228 geometric departure of the real enclosure, with its harness penetrations, from  
 229 the parallel-plate idealisation with residual agglomeration error. All loads  
 230 below are the row-normalised values; the unnormalised production values  
 231 are kept as an upper band. Normalisation enforces closure, not reciprocity  
 232  $A_i F_{ij} = A_j F_{ji}$ ; the least-squares rectification procedures compared by Taylor  
 233 and Luck [44] enforce both and would be the next refinement. Two residuals

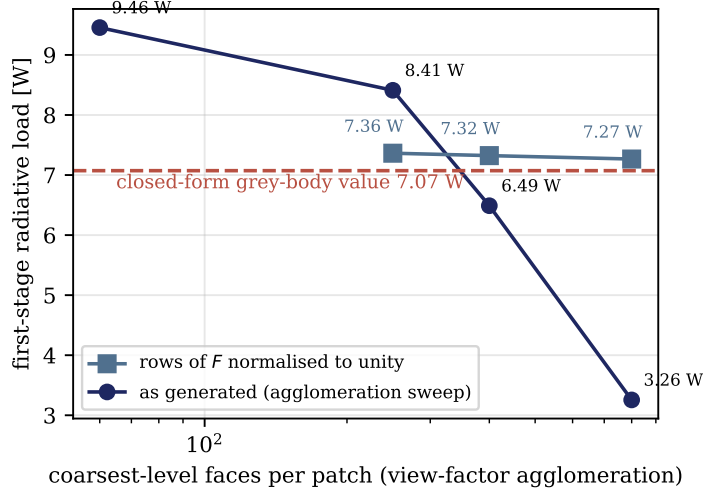


Figure 5: First-stage radiative load (300  $\rightarrow$  50 K) against the view-factor agglomeration level with the matrix as generated (circles) and with its rows normalised to unity (squares), compared with the closed-form grey-body value. The monotonic collapse of the generated matrix under refinement is a closure artefact amplified by the low emissivities; the normalised load converges toward the closed form.

quantify what it leaves: the net radiative power summed over every surface of a closed region, which must vanish, is 1.5 W for the generated matrix at 400 faces and 14 W at 800, against 0.10, 0.13 and 0.36 W (1.3, 1.8 and 4.9 % of the load) for the normalised matrices at 250, 400 and 800 faces; and the area-weighted reciprocity error  $\sum |A_i F_{ij} - A_j F_{ji}| / \sum A_i F_{ij}$ , with coarse-face areas rebuilt from the mesh (`reciprocity.py`), is 3.9, 5.5 and 12.6 % at the same levels (outer gap region). Both grow with refinement, which is why the production level is 250 faces and why the 3–4 % gap to the closed form is reported as the accuracy of the method.

### 3.2. Stage loads and mechanism split

At the first stage radiation dominates and the shield temperature barely matters (Table 3, Fig. 6). The radiative flux scales with  $T_{\text{hot}}^4 - T_{\text{cold}}^4$  and  $300^4$  dwarfs both  $50^4$  and  $77^4$ , so the 77 K shield intercepts 99.6 % of what the 50 K shield intercepts. Harness conduction falls with the smaller span, 962 against 1029 mW. The first-stage loads are 8.39 W (50 K) and 8.30 W (77 K); the emissivity band of  $\pm 50$  % moves them between 6.5 and 12.5 W but leaves

Table 3: Stage loads by mechanism (row-normalised view factors; closed-form value in parentheses), with the unnormalised production value and the  $\pm 50\%$  emissivity band as upper and lower bounds.

| Case                          | Radiation      | Conduction       | Total    | Prod.    | $\varepsilon \pm 50\%$<br>[W] |
|-------------------------------|----------------|------------------|----------|----------|-------------------------------|
| 300 $\rightarrow$ 50 K        | 7.36 W (7.07)  | 1.029 W (1.029)  | 8.39 W   | 9.44 W   | 6.5–12.5                      |
| 300 $\rightarrow$ 77 K        | 7.34 W (7.04)  | 0.962 W (0.962)  | 8.30 W   | 9.35 W   | 6.4–12.4                      |
| 50 $\rightarrow$ 4 K          | 3.16 mW (3.15) | 70.3 mW (70.3)   | 73.5 mW  | 75.6 mW  | –                             |
| 77 $\rightarrow$ 4 K          | 17.8 mW (17.7) | 162.1 mW (162.1) | 179.9 mW | 191.8 mW | –                             |
| 300 $\rightarrow$ 50 K<br>MLI | 1.69 W (1.47)  | 1.029 W          | 2.72 W   | 2.84 W   | –                             |
| 300 $\rightarrow$ 77 K<br>MLI | 1.68 W (1.46)  | 0.962 W          | 2.65 W   | 2.77 W   | –                             |

the 1% difference between routes unchanged. The conservative production value of 9.4 W is used as the cold-plate duty in Section 4.

At the 4 K stage the picture inverts. Residual radiation rises by the factor  $(77^4 - 4^4)/(50^4 - 4^4) = 5.6$ , but from a base of 3 mW; harness conduction rises by the factor 2.31 from a base of 70 mW. The 77 K route therefore delivers 106 mW more to the 4 K plate: 92 mW of conduction and 15 mW of radiation, 86% conductive. Through the ideal 4 K multiplier of 74 this is 7.9 W of room-temperature power, 6.8 W of it from the harness. Every stage-integrated ratio agrees with its closed form to better than 0.5%.

The MLI-equivalent shield ( $\varepsilon = 0.003$  on the first-stage surfaces) cuts the first-stage radiation to 1.69 W (50 K) and 1.68 W (77 K) with the row-normalised matrix (closed form 1.47 and 1.46 W; the unnormalised production matrix gives 1.81 W on either route, the same +23% closure bias as at the bare-shield level), so the first-stage loads fall to 2.72 and 2.65 W, with conduction's share rising to 36–38%. The 4 K-stage loads are unaffected. This variant is the one that changes the ledger verdict in Section 5.

### 3.3. Spatial distribution

The view-factor solution concentrates the radiative flux on the shield regions with a direct line of sight to the OVC (8.2–10.8 W m<sup>-2</sup> across the shield) and leaves the harness penetrations cooler (Fig. 7). Conduction is shared among the bundles in the ratio of their line counts: on the 77  $\rightarrow$  4 K span the drive and flux families deliver 125 of the 162 mW. These two bundles are the targets for any intermediate heat sink, and the harness conduction

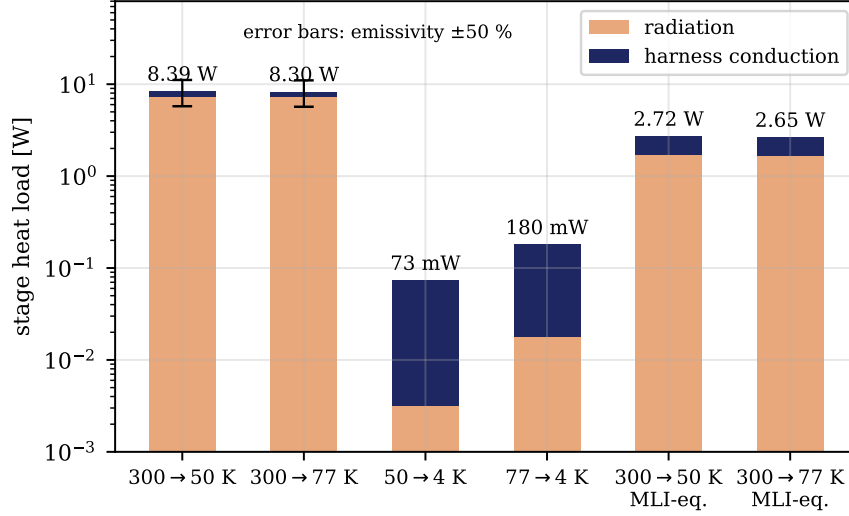


Figure 6: Stage loads for the four enclosure cases and the two MLI-equivalent variants (row-normalised view factors, the values used in the text), decomposed by mechanism, with the  $\pm 50\%$  emissivity band. The first stage is radiation-dominated and nearly identical between routes; the 4 K stage is conduction-dominated and is where the routes diverge.

integral, not the cryogen, is the lever on the 4 K-stage penalty.

## 4. Results Q2: helium gas versus liquid nitrogen in the cold plate

### 4.1. Verification

Every forced-convection run closes its energy balance, wall heat input against mass-flux-weighted enthalpy rise, to within 0.2% (Supplementary Table S7). At  $Re = 500$  the helium flow is thermally developing over the first 59% of the channel ( $L_t/L = 0.59$ ) and returns  $Nu = 3.66$ , 35% above the fully developed limit of 2.71 for a square duct heated on one wall [41], the excess being the developing-length contribution; every other laminar run has a longer development length ( $L_t/L$  up to 8.8 for nitrogen) and lies further above the limit, as it must. The three-grid studies (Table 4) converge monotonically for both fluids: the production-mesh grid convergence index is 0.7% ( $Nu$ ) and 2.5% ( $f$ ) for helium and 2.2% and 1.6% for nitrogen. In the turbulent runs the first-cell  $y^+$  on the heated wall averages 2.5–5.6 with a maximum of 9.5, within the low-Reynolds treatment of the SST model.

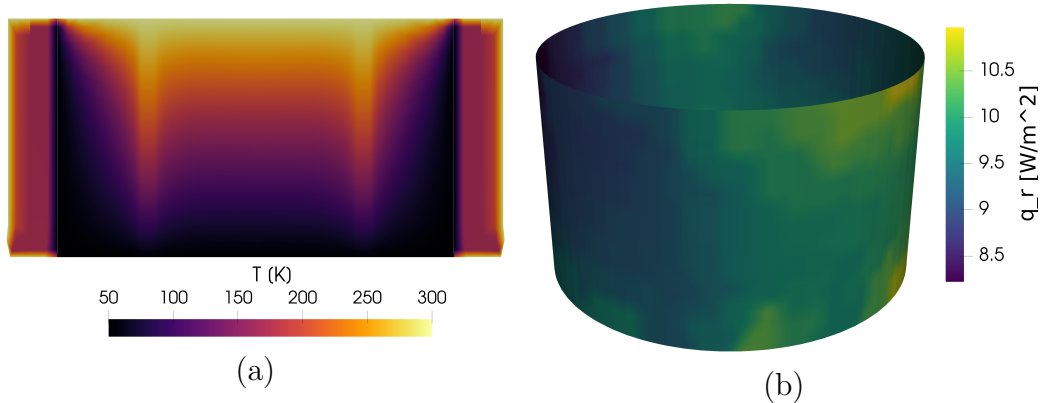


Figure 7: (a) Mid-plane temperature and (b) shield radiative-flux map of the 300 → 50 K case.

288 Nitrogen’s turbulent Nusselt numbers lie 8–9 % below the Gnielinski correla-  
 289 tion [42] and helium’s 28–32 % below it. The correlation is for a duct heated  
 290 on all walls; in the laminar limit one heated wall lowers  $Nu$  by 25 % (2.71  
 291 against 3.61), and the larger deficit of the low-Prandtl gas, whose thermal  
 292 layer fills the duct, is consistent with that boundary condition but is not  
 293 separately verified here, so the correlation is a reference, not a benchmark,  
 294 for these runs.

295 *Property sensitivity.* Two helium runs at 1 bar,  $Re = 500$  and 2300, were  
 296 repeated with cubic  $\mu(T)$  and  $k(T)$  fitted to the NIST data over 40–100 K  
 297 (Section 2.3) and a heat-flux wall condition using the local conductivity (de-  
 298 livered 9.399 W, closure 0.01 %). Because  $k$  rises with temperature, the hot  
 299 near-wall gas conducts better than at the 50 K reference: at  $Re = 500$  the  
 300 Nusselt number rises from 3.66 to 4.16 (+14 %), the plate mean falls from  
 301 87.4 to 83.0 K and the hottest face from 106.1 to 97.3 K, while the friction  
 302 factor and pumping power rise by 5 %; at  $Re = 2300$  the changes are +5 %  
 303 in  $Nu$ ,  $-1.0$  K in the mean and  $-2.9$  K in the hottest face (79.7 to 76.9 K).  
 304 The 0.5 K by which that face clears 77.4 K is within the uncertainty of the  
 305 quantity (the three-grid study moves it by 0.17 K; the property fits and con-  
 306 stant  $c_p$  add a few tenths), so the hottest-face optimum is stated as a range,  
 307  $Re \approx 2300$  to 5000 and a pumping-power ratio of 4 to 30. The constant-  
 308 property results are conservative toward helium by up to 14 % in  $h$  at the  
 309 lowest Reynolds number and 5 % at the design point; the rankings below are  
 310 not affected. Both runs are listed in Supplementary Table S7.

Table 4: Grid convergence at  $Re = 2300$  (Celik et al. [43]; meshes of 86k, 292k and 984k cells, refinement ratio 1.5). The production mesh is the medium one.

| Fluid                   | Quantity | Coarse | Medium | Fine   | $p$  | Extrapolated | $GCI_{\text{medium}}$ |
|-------------------------|----------|--------|--------|--------|------|--------------|-----------------------|
| He (1 bar)              | $Nu$     | 6.439  | 6.449  | 6.457  | 0.64 | 6.49         | 0.7 %                 |
|                         | $f$      | 0.0632 | 0.0642 | 0.0647 | 1.40 | 0.0655       | 2.5 %                 |
| LN <sub>2</sub> (3 bar) | $Nu$     | 10.51  | 10.44  | 10.38  | 0.87 | 10.26        | 2.2 %                 |
|                         | $f$      | 0.0592 | 0.0599 | 0.0603 | 1.71 | 0.0607       | 1.6 %                 |

#### 311 4.2. Operating maps

312 Figure 8 maps the two coolants over  $Re = 500\text{--}10^4$  at the fixed 9.4 W  
313 duty. Nitrogen’s heat-transfer coefficient exceeds helium’s by 4.9–5.1 times  
314 in the laminar range and 6.2–6.3 in the turbulent range; the laminar ratio  
315 is the conductivity ratio of 3.1 times the ratio of developing-flow Nusselt  
316 numbers, which favours the high-Prandtl liquid. The consequence for the  
317 plate is stark. At atmospheric pressure the helium-cooled wall averages 87 K  
318 at  $Re = 500$  and 67 K at  $Re = 2300$ , with hottest faces at the channel outlet  
319 of 106 and 80 K (Fig. 8d). The plate mean falls below 77 K above  $Re \approx 970$ ,  
320 the hottest face only in turbulent flow at  $Re = 5000$  (61 K maximum, 55 K  
321 mean), and the nominal 50 K is approached only at  $Re = 10^4$  and 65 mW of  
322 pumping power (50.3 K mean, 54.3 K maximum). The nitrogen-cooled wall  
323 stays between 78 and 85 K in the mean and below 89.2 K at its hottest face;  
324 its saturation margin at 5 bar is 4.8 K at  $Re = 500$ , 9.9 K at 2300 and 15.4 K  
325 at  $10^4$ . At 3 bar the  $Re = 500$  run would exceed saturation locally (89.2  
326 against 87.9 K), and at 1 bar every run would exceed 77.4 K, so a pressurised  
327 or subcooled loop is a requirement, not a refinement.

328 Pumping power separates the fluids in the opposite sense once pressure  
329 enters. At 1 bar helium needs 38 times nitrogen’s pumping power at matched  
330 Reynolds number (1.39 against 0.037 mW at  $Re = 2300$ ), because its mass  
331 flow is 26 times smaller but its velocity 32 times larger. Raising the helium  
332 pressure at fixed Reynolds number leaves the mass flow, the Nusselt number  
333 and the wall temperature unchanged to four significant figures and reduces  
334 the pumping power as the inverse square of the density: 25.0 times at 5 bar  
335 and 324 times at 18 bar (Fig. 8c). At the 18 bar of the ITER and DTT shield  
336 loops [4, 5] helium’s pumping power is 4.3  $\mu\text{W}$ , 8.5 times below nitrogen’s; the  
337 parallel-path gaseous-helium shields of LHD operate in the same regime [6].

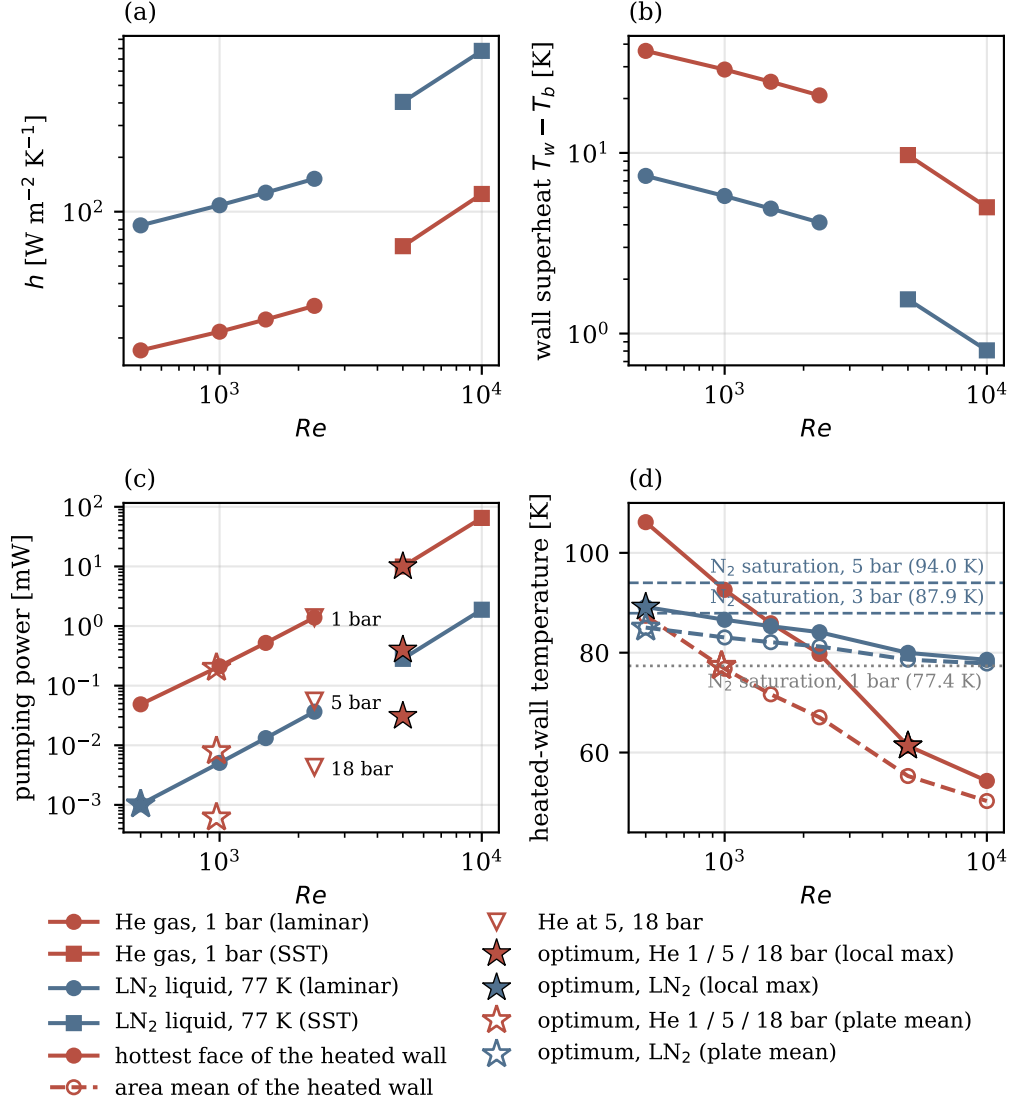


Figure 8: Operating maps at 9.4 W duty: (a) heat-transfer coefficient, (b) wall superheat, (c) pumping power with the helium pressure series at  $Re = 2300$ , and (d) heated-wall temperature, hottest face (filled) and area mean (open), with the nitrogen saturation temperatures at 1, 3 and 5 bar. Circles: laminar; squares:  $k-\omega$  SST; stars: each fluid's optimum, the minimum pumping power at which the wall temperature meets its limit (77.4 K for helium, 92.0 K for nitrogen), filled for the hottest-face criterion and open for the plate-mean criterion.

338 *Optimum operating points.* The second half of Q2 needs an optimum: the  
 339 minimum pumping power at which the plate meets a wall-temperature limit,  
 340 77.4 K for helium (the temperature a nitrogen-bath plate would hold, be-  
 341 yond which the 50/77 K distinction is forfeited) and the 5 bar saturation  
 342 temperature less a 2 K margin, 92.0 K, for nitrogen. The answer depends on  
 343 which wall temperature is held to the limit. If the hottest face must com-  
 344 ply, no laminar helium run of the constant-property sweep does: the 1 bar  
 345 curve of Fig. 8d stays above 79.7 K up to  $Re = 2300$ , and the first com-  
 346 pliant run is the turbulent  $Re = 5000$  case at 9.95 mW (1 bar), 0.40 mW  
 347 (5 bar) and 31  $\mu$ W (18 bar); the laminar–turbulent gap is not interpolated.  
 348 The 2.4 K margin at  $Re = 2300$  is closed by the variable-property run of Sec-  
 349 tion 4.1 (76.9 K), which would move the optimum to  $Re \approx 2300$  at 1.39 mW  
 350 or 4.3  $\mu$ W at 18 bar. Nitrogen complies at  $Re = 500$ , the lowest run, with  
 351 1.0  $\mu$ W. If the plate mean is the criterion, helium crosses 77.4 K at  $Re \approx 970$   
 352 (0.20 mW at 1 bar, 0.62  $\mu$ W at 18 bar) and nitrogen again at  $Re = 500$  (stars  
 353 in Fig. 8c,d). At their optima, pressurised helium therefore needs 4 to 30  
 354 times nitrogen’s pumping power on the hottest-face criterion and 40 % less  
 355 on the plate-mean criterion. Both are microwatts and neither enters the  
 356 ledger; what the comparison establishes is that the 38-fold penalty of atmo-  
 357 spheric helium at matched  $Re$  is the cost of the wrong loop pressure, not  
 358 a property of the gas, and that a helium plate must be checked against its  
 359 hottest face, 13 K above the mean at  $Re = 2300$ .

### 360 4.3. Wall-temperature distributions

361 Figure 9 gives the heated-wall temperature along the centre row of the  
 362 first channel, read from the wall faces themselves, and the cross-channel  
 363 profile at mid-length. Along the wall the temperature rise grows as  $x^{1/3}$  in  
 364 the developing region and then linearly; at the outlet the helium wall sits  
 365 30 K above its inlet at  $Re = 2300$  and 58 K at  $Re = 500$ , nitrogen 5.5 and  
 366 10.5 K, and the hottest faces of the patch, near the duct side walls at the  
 367 outlet where the velocity is lowest, are a further 4.8 K (helium) and 1.6 K  
 368 (nitrogen) warmer at  $Re = 2300$ . Across the channel the helium thermal  
 369 layer at  $Re = 500$  fills the duct, whereas nitrogen’s occupies the near-wall  
 370 30 % even at  $Re = 500$ . This is the geometric origin of the heat-transfer ratio  
 371 and of the plate temperatures in Fig. 8d.

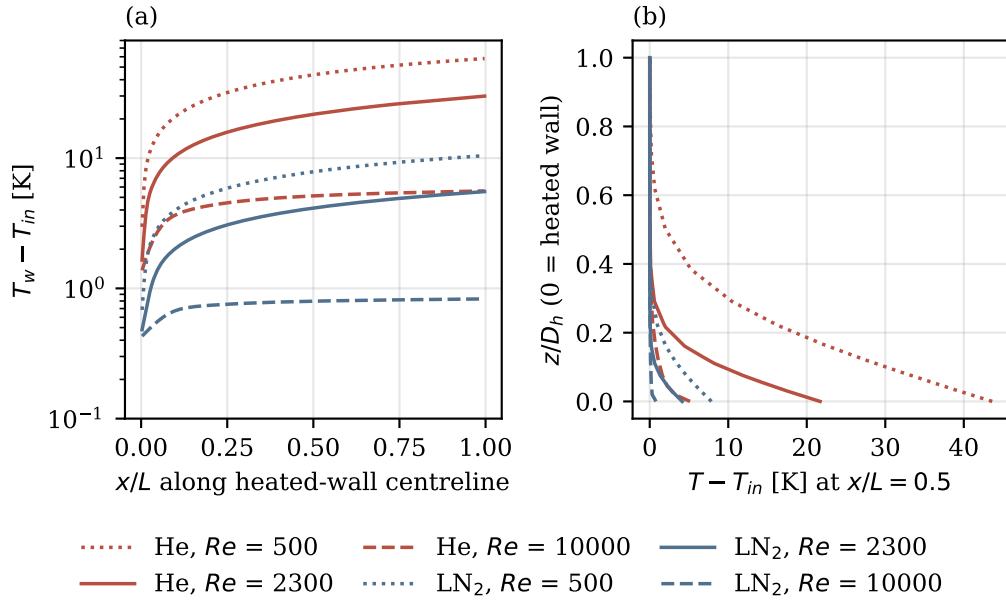


Figure 9: (a) Heated-wall centreline temperature rise above the inlet and (b) cross-channel temperature profile at  $x/L = 0.5$  for both fluids at  $Re = 500$ ,  $2300$  and  $10^4$ .

Table 5: Design point,  $Re = 2300$ , 9.4 W, forced convection. He at 1 bar (18 bar in parentheses where pressure matters); LN<sub>2</sub> at 5 bar (3 bar in parentheses for the saturation margin). GCI on  $Nu$  and  $h$ : 0.7 % (He), 2.2 % (LN<sub>2</sub>).

| Quantity                                                           | He, 50 K      | LN <sub>2</sub> , 77 K | LN <sub>2</sub> /He |
|--------------------------------------------------------------------|---------------|------------------------|---------------------|
| Mass flow $\dot{m}$ [g s <sup>-1</sup> ]                           | 0.731         | 18.8                   | 25.7                |
| Volumetric flow [cm <sup>3</sup> s <sup>-1</sup> ]                 | 702 (39)      | 23.3                   | 0.033               |
| Mean velocity $U$ [m s <sup>-1</sup> ]                             | 1.52 (0.084)  | 0.047                  | 0.031               |
| Inlet / outlet temperature [K]                                     | 45.0 / 47.5   | 77.0 / 77.2            | –                   |
| Pressure drop $\Delta p$ [Pa]                                      | 1.98 (0.11)   | 1.57                   | 0.80                |
| Pumping power $\dot{W}_p$ [ $\mu$ W]                               | 1388 (4.3)    | 36.6                   | 0.026 (8.5)         |
| Wall temperature, mean / hottest face <sup>a</sup> [K]             | 67.1 / 79.7   | 81.2 / 84.1            | –                   |
| Wall superheat $T_w - T_b$ [K]                                     | 20.8          | 4.12                   | 0.20                |
| Heat-transfer coefficient $h$ [W m <sup>-2</sup> K <sup>-1</sup> ] | 30.1          | 152                    | 5.05                |
| Nusselt number $Nu$                                                | 6.45          | 10.4                   | 1.62                |
| Darcy friction factor $f$                                          | 0.064         | 0.060                  | 0.93                |
| NTU / effectiveness $\varepsilon$                                  | 0.119 / 0.112 | 0.060 / 0.058          | 0.50 / 0.52         |
| Mouromtseff number (SI units)                                      | 567 (5720)    | $4.6 \times 10^4$      | 81 (8)              |
| Buoyancy parameter $Gr/Re^2$                                       | 0.02          | 1.05                   | –                   |
| Saturation margin $T_{sat} - T_{w,max}$ [K]                        | –             | 9.9 (3.8)              | –                   |
| Energy-balance closure                                             | 0.06 %        | 0.001 %                | –                   |

<sup>a</sup>patch area mean / hottest patch face (`surfaceFieldValue`), at the outlet end.

#### 372 4.4. Ranking on three bases

373 Table 5 collects the design point and Fig. 10 the trade-off between wall  
374 superheat and pumping power. At matched Reynolds number nitrogen wins  
375 heat transfer by a factor of 5 and pumping power by 38; the effectiveness–  
376 NTU pair favours helium (0.11 against 0.06), a capacity-rate artefact of the  
377 gas’s small  $\dot{m}c_p$  that says nothing about how cold the plate is. At matched  
378 mass flow nitrogen’s Reynolds number would fall to about 90, below the  
379 sweep; the one-wall limit  $Nu_\infty \approx 2.71$  bounds its heat-transfer coefficient  
380 from below at about 40 W m<sup>-2</sup>K<sup>-1</sup>, 1.3 times helium’s at  $Re = 2300$ , quoted  
381 only as a bound. At matched pumping power, the basis a plant designer  
382 pays for, nitrogen at the pumping power of 1 bar helium ( $Re = 2300$ ) holds  
383 a superheat of 0.9 K against helium’s 20.8 K, a factor of 23; at that of 18 bar  
384 helium the factor is 3.5 (5.9 against 20.8 K). In the turbulent range the curves  
385 of Fig. 10 are parallel: a given superheat costs helium at 1 bar roughly 30  
386 times nitrogen’s pumping power, and the 18 bar helium point sits inside the  
387 nitrogen envelope on pumping power but outside it on superheat. No single  
388 number ranks the coolants; the operating point and the pressure do.

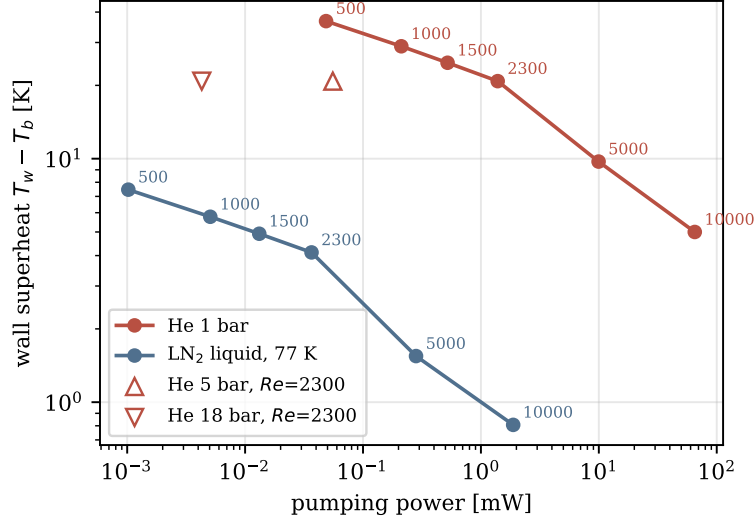


Figure 10: Wall superheat against pumping power for both fluids over the Reynolds sweep (labels:  $Re$ ), with the helium pressure series at  $Re = 2300$ .

#### 4.5. Buoyancy sensitivity

The buoyancy parameter  $Gr/Re^2$  of the forced-convection runs is at most 0.63 for helium at 1 bar ( $Re = 500$ ) and above unity for nitrogen in the laminar range (40 at  $Re = 500$ , 1.0 at 2300), so a real plate in which gravity acts along the heated wall would carry a mixed-convection secondary flow in nitrogen; for helium only the dense 18 bar loop is affected ( $Gr/Re^2 = 7.3$  at  $Re = 2300$ ). The four gravity-on runs confirm this. Helium converged and gained 27% in  $Nu$  at  $Re = 2300$  and 118% at  $Re = 500$ . Nitrogen reached no steady state at either Reynolds number: after 10 000 iterations the temperature residual oscillated at  $10^{-3}$  and the energy balance stayed open by 11% and 72%, the signature of an unsteady buoyant flow a steady solver cannot represent. The forced-convection nitrogen results therefore describe a plate with gravity normal to the heated wall or absent; with gravity along the wall the steady model gives no bound in either direction, and the ranking of Section 4.4 is stated for forced convection only.

### 5. Results Q3: the work ledger

Table 7 and Fig. 11 price the two routes with Eq. (7). With ideal multipliers and the bare shield the nitrogen route is cheaper: 37.3 against 47.4 W,

407 a margin of 21 %, because intercepting 8.3 W at a multiplier of 2.9 instead  
 408 of 5.0 saves 18 W while the 4 K-stage penalty costs 7.9 W. The emissivity  
 409 band (31.8–49.2 W against 37.8–68.0 W) does not close the gap. With the  
 410 MLI-equivalent shield the first-stage load shrinks to 2.6–2.7 W, the nitrogen  
 411 saving to 5.9 W, and the verdict flips: helium 19.0 W against nitrogen 21.0 W.  
 412 The ideal ledger therefore favours nitrogen where the first-stage load is large  
 413 and helium where insulation has removed it, which is the regime of the MLI-  
 414 wrapped shields surveyed by Shu et al. [3] and of the single-layer intercepts  
 415 of Singh et al. [45].

416 With hardware efficiencies, and the shields at their nominal temperatures,  
 417 the verdict is not close in either case. The 4 K stage at 0.8–1.5 % of Carnot  
 418 turns the 106 mW penalty into 520–980 W, more than the whole first-stage  
 419 bill of either route; the helium route costs 840 W (640–1100) against 1300 W  
 420 (960–1770) for nitrogen with the bare shield and 610 against 1240 W with  
 421 MLI. The nitrogen first-stage term is not a cryocooler but the liquefaction  
 422 energy of the liquid evaporated, 0.5–0.7 kWh kg<sup>-1</sup> against a latent heat of  
 423 199 kJ kg<sup>-1</sup>, or  $M_{N_2} = 9.1$ –12.7 W of plant input per watt intercepted (a  
 424 fraction-of-Carnot basis for an 80 K cooler, 7.2–9.7, would lower the nitrogen  
 425 figures by 1–2 %); the 4 K-stage term is the same cryocooler stage on both  
 426 routes. Written as a break-even, the nitrogen route is cheaper only if

$$\frac{\eta_2}{\eta_1} \gtrsim \frac{74 \Delta \dot{Q}_2}{5.0 \dot{Q}_{1,\text{He}} - M_{N_2} \dot{Q}_{1,N_2} \eta_1}, \quad (8)$$

427 which with the present loads and the midpoints  $\eta_1 = 0.12$  and  $M_{N_2} = 10.7$   
 428 requires a 4 K-stage efficiency of 3.0 % of Carnot for the bare shield and 9 %  
 429 for the MLI shield, two to ten times what commercial coolers achieve [30].  
 430 The lever is  $\Delta \dot{Q}_2$ : because 86 % of it is harness conduction, a harness that  
 431 cuts the 77 → 4 K conduction penalty threefold, by heat-sinking the drive and  
 432 flux bundles or by lower-conductivity outer conductors, brings the bare-shield  
 433 break-even to 1.3 % of Carnot, within the range of real hardware.

434 *Coupled operating points.* The ledger above holds each shield at its nominal  
 435 temperature, but Section 4.2 shows that neither cold plate sits there: the  
 436 helium plate averages 67 K at  $Re = 2300$  and reaches 50 K only at  $Re = 10^4$ ;  
 437 the nitrogen plate averages 81.2 K at  $Re = 2300$  and 77.8 K at  $10^4$ , 4.2 and  
 438 0.8 K above the bath, because the duty must cross the wall superheat. The  
 439 harness is heat-sunk at the plate, so the 4 K-stage load follows the plate tem-  
 440 perature. Table 6 re-evaluates the ledger with the first-stage temperature set

Table 6: Work ledger at coupled operating points: first-stage temperature  $T_1$  equal to the area-mean plate temperature of the cold-plate model (He at 1 bar; 18 bar changes only the negligible pumping term);  $\dot{Q}_2$  from the harness integral and the scaled residual radiation; equivalent room-temperature power with ideal multipliers and with mid-range hardware efficiencies (ranges in `data/ledger_coupled.csv`).

| Operating point (He / LN <sub>2</sub> )                         | $T_1$ [K]   | $\dot{Q}_2$ [mW] | $\dot{W}_{\text{ideal}}$ [W] | $\dot{W}_{\text{real}}$ [W] | He advantage, |
|-----------------------------------------------------------------|-------------|------------------|------------------------------|-----------------------------|---------------|
| <i>bare shield</i>                                              |             |                  |                              |                             |               |
| nominal, decoupled                                              | 50 / 77     | 73 / 180         | 47.4 / 37.3                  | 840 / 1300                  | 3             |
| matched $Re = 2300$ , both laminar                              | 67.1 / 81.2 | 135 / 201        | 51.7 / 38.8                  | 1250 / 1450                 | 1             |
| own design points: He $Re = 10^4$ , LN <sub>2</sub> $Re = 2300$ | 50.3 / 81.2 | 74 / 201         | 47.5 / 38.8                  | 850 / 1450                  | 4             |
| both turbulent, $Re = 10^4$                                     | 50.3 / 77.8 | 74 / 184         | 47.5 / 37.6                  | 850 / 1330                  | 3             |
| <i>MLI-equivalent shield</i>                                    |             |                  |                              |                             |               |
| nominal, decoupled                                              | 50 / 77     | 73 / 180         | 19.0 / 21.0                  | 610 / 1240                  | 5             |
| matched $Re = 2300$ , both laminar                              | 67.1 / 81.2 | 135 / 201        | 23.4 / 22.5                  | 1020 / 1380                 | 2             |
| own design points                                               | 50.3 / 81.2 | 74 / 201         | 19.1 / 22.5                  | 610 / 1380                  | 5             |
| both turbulent, $Re = 10^4$                                     | 50.3 / 77.8 | 74 / 184         | 19.1 / 21.3                  | 610 / 1270                  | 5             |

441 to the area-mean plate temperature at four operating points, using the har-  
 442 ness integral behind Eq. (6) and the  $T_1^4$  scaling of the residual radiation, both  
 443 reproduced by the enclosure model to 0.5 %, with the helium pumping power  
 444 charged at a 50 % circulator efficiency. At matched laminar flow the helium  
 445 plate's 67 K costs 62 mW at the 4 K stage and the nitrogen plate's 81 K costs  
 446 21 mW, and helium's hardware advantage shrinks from 36 to 13 % (bare) and  
 447 from 51 to 26 % (MLI). With each route at the flow its design implies, helium  
 448 turbulent at  $Re = 10^4$  to hold 50 K and nitrogen laminar at  $Re = 2300$ , the  
 449 advantage is 41 and 56 %; with both turbulent, 36 and 52 %. The ideal ledger  
 450 moves the other way: at matched laminar flow nitrogen is cheaper by 25 %  
 451 (bare) and 4 % (MLI). These are equivalent room-temperature powers under  
 452 the stated efficiencies and supply energies, not measured consumption; they  
 453 establish the sign and order of the difference, and that helium's advantage  
 454 exists only where its plate is actually held near 50 K, which the flow rate, not  
 455 the coolant, decides.

## 456 6. Discussion

457 *The penalty is a harness property.* The passive cost of the warmer shield is  
 458 located, sized and attributed: 106 mW at the 4 K stage, 86 % of it conduction  
 459 along a harness whose drive and flux families carry three quarters. It is  
 460 a property of the wiring, not of the cryogen, and Eq. (8) shows that, at

Table 7: Work ledger at the nominal shield temperatures: stage loads (row-normalised view factors) and equivalent room-temperature power, ideal and hardware-based. Ranges: emissivity  $\pm 50\%$  (ideal) and the stage-efficiency and nitrogen supply-energy ranges of Supplementary Table S9 (real).

| Shield  | Route                  | $\dot{Q}_1$ [W] | $\dot{Q}_2$ [mW] | $\dot{W}_{\text{ideal}}$ [W] | $\dot{W}_{\text{real}}$ [W] |
|---------|------------------------|-----------------|------------------|------------------------------|-----------------------------|
| bare    | He, 50 K               | 8.39            | 73.5             | 47.4 (37.8–68.0)             | 840 (640–1100)              |
| bare    | LN <sub>2</sub> , 77 K | 8.30            | 179.9            | 37.3 (31.8–49.2)             | 1300 (960–1770)             |
| MLI-eq. | He, 50 K               | 2.72            | 73.5             | 19.0                         | 610 (450–820)               |
| MLI-eq. | LN <sub>2</sub> , 77 K | 2.65            | 179.9            | 21.0                         | 1240 (910–1700)             |

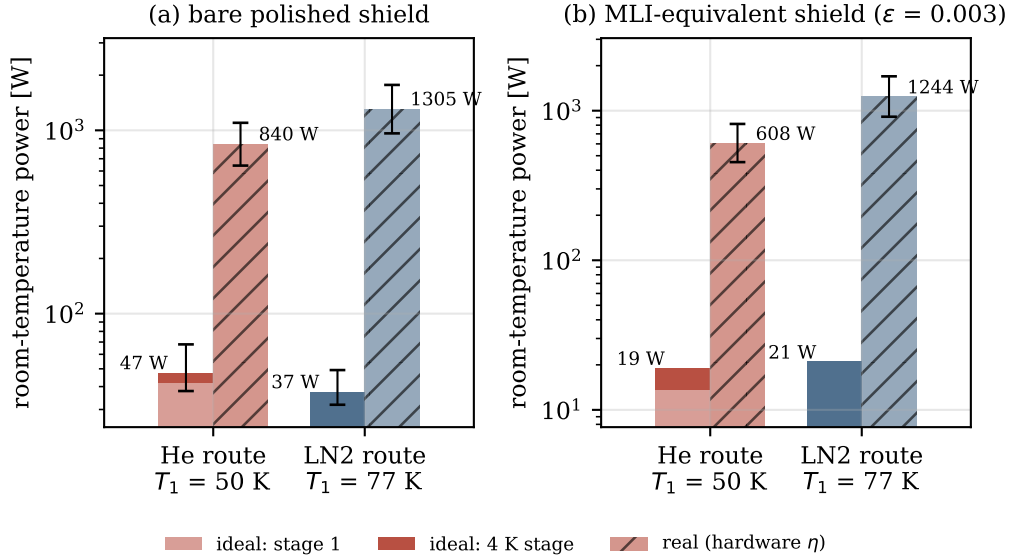


Figure 11: Room-temperature power of the two routes for (a) the bare polished shield and (b) the MLI-equivalent shield: ideal Carnot multipliers split by stage (solid) and hardware efficiencies (hatched), with emissivity and efficiency ranges as error bars.

the efficiencies of present hardware, the wiring is the variable that moves the break-even most. Lumped budgets give the same total, and its split among bundles follows from the cable counts; what the resolved model adds is the coupling of the harness to the radiation field, the flux distribution over the shield, the closure behaviour of the view-factor matrix, the confirmation that the residual radiation is a minor term at any shield temperature, and the dependence of the 4 K load on the plate temperature that the coupled ledger of Table 6 uses. This is the design steer that Li et al.’s optimisation of intercepts on supports and in insulation [11] cannot give for a wired cryostat.

*The helium plate floats..* A cold plate cooled by circulated helium gas at atmospheric pressure does not sit at the gas temperature. At the 9.4 W duty it runs 21 K above the bulk at  $Re = 2300$ , with a hottest face 13 K above that mean; the mean reaches nitrogen temperature at  $Re \approx 970$  and the hottest face only at  $Re \gtrsim 2300$ , and the nominal 50 K is approached only at  $Re = 10^4$  and tens of milliwatts of pumping power. Pressurising the loop leaves the wall temperature untouched but makes the pumping power negligible. This is a statement about the circulated-gas architecture and the channel sizing, not about the conductively coupled PTR first stage of Fig. 1a, whose plate temperature is set by the cooler’s capacity curve. It does mean that the 50/77 K distinction of the passive comparison is only realised on the helium route with an adequate flow or pressure, and Section 4.2 gives the conditions.

*The three questions couple through the plate temperature..* The ledger credits the helium route with a 50 K shield, but the cold plate reaches 50 K only at  $Re \approx 10^4$ ; at the laminar design point of  $Re = 2300$  and 1 bar its mean is 67 K, where the harness integral of Eq. (6) gives a 4 K-stage conduction of 125 mW against 70 mW at 50 K and 162 mW at 77 K. A helium shield run there forfeits 54 mW, 59 % of the 92 mW conduction penalty that separates the routes, and about 4 W of the 7.9 W ideal advantage; the 65 mW of pumping power that buys the 50 K plate at  $Re = 10^4$  is negligible against that, so on the helium route the flow rate is set by the 4 K ledger. The nitrogen plate is not pinned by the bath either: it sits 4.2 K above it at  $Re = 2300$  and 0.8 K at  $10^4$ , worth 21 and 4 mW at the 4 K stage, a smaller and flow-insensitive coupling because the liquid’s superheat is small. Table 6 carries both into the ledger; what remains of the bath-pinning idea is a plate temperature that the flow rate barely moves.

496 *The basis of comparison decides the ranking..* Matched Reynolds number,  
 497 matched mass flow and matched pumping power rank the fluids differently  
 498 in magnitude and, for pumping power at 18 bar, in sign. The effectiveness–  
 499 NTU pair inverts the engineering conclusion because it rewards a small ca-  
 500 pacity rate. A cryogenic coolant comparison must therefore state its basis  
 501 and its operating point, as Gačnik et al. found necessary for nitrogen-cooled  
 502 terminals [29]; the operating maps of Fig. 8 are offered as the form such a  
 503 comparison should take.

504 *Ideal and real ledgers disagree, and the disagreement is informative..* The  
 505 ideal ledger prefers nitrogen for a bare shield and helium for an insulated  
 506 one; the hardware ledger prefers helium in both cases, by 36–51 % at the  
 507 nominal shield temperatures and by 13–56 % at the coupled operating points  
 508 of Table 6, because the 4 K stage of a commercial cooler converts each milli-  
 509 watt of penalty into 5–9 W at the wall. The nitrogen route retains what Li et  
 510 al. identified three decades ago [11]: a constant shield temperature, stored re-  
 511 serve, and independence from the cryocooler, which now also means freedom  
 512 from its vibration [46]. Those are operational arguments; the thermodynamic  
 513 one is conditional on the harness.

514 *Methodological findings..* Two travel beyond this geometry. First, view-  
 515 factor matrices generated for low-emissivity enclosures must be checked for  
 516 closure and, if the enclosure is closed, row-normalised, which restores closure  
 517 though not reciprocity; the required closure accuracy is of order the emissiv-  
 518 ity, and agglomeration refinement does not supply it, whereas the normalised  
 519 load converges under it. The benchmark practice of Končar et al. [25] should  
 520 be extended by this test. Second, the mixing-cup temperature on a graded  
 521 mesh must be mass-flux weighted without an additional area factor; the cor-  
 522 rected accounting closed the energy balance of every run from a few per cent  
 523 to below 0.2 %.

524 *Limitations..* The enclosure imposes fixed temperatures on its plates and  
 525 shield and neglects contact resistance at the harness ends; a finitely conduc-  
 526 tive shield would carry the radiative-flux map of Fig. 7 into a temperature  
 527 map, which is the next step. The cold plate is a single-wall-heated smooth  
 528 duct with constant properties except for the helium density and the nitrogen  
 529 buoyancy term; conjugate conduction in the plate, which would spread the  
 530 wall temperature, is not resolved. Nitrogen is single phase by verified mar-  
 531 gin at 3 bar but not at 1 bar. The turbulent runs rely on the SST model at

532  $y^+ \leq 9.5$  and were not compared with experiment. The hardware efficiencies  
533 are survey ranges, with the two-stage cooler's efficiency split between stages  
534 from those ranges rather than from one machine; a measured capacity map  
535 would replace them. Nitrogen buoyancy was shown to be unsteady and ex-  
536 cluded from the ranking rather than resolved; a transient calculation in a  
537 specified plate orientation would quantify it.

## 538 7. Conclusions

- 539 1. A 77 K first shield costs essentially nothing at the first stage relative to  
540 a 50 K shield: 99.6 % of the radiation and 6.5 % less harness conduction,  
541 8.3 against 8.4 W. The penalty appears at the 4 K stage, 106 mW, 86 %  
542 of it harness conduction in the drive and flux bundles, worth 7.9 W of  
543 ideal room-temperature power and 520–980 W with real 4 K coolers.
- 544 2. View-factor radiation in low-emissivity enclosures is controlled by ma-  
545 trix closure: the generated matrix gave 9.5 to 3.3 W under agglomera-  
546 tion refinement against a 7.07 W closed form; row normalisation recov-  
547 ers it to 4 % at the first stage and 0.5 % at the 4 K stage and converges  
548 under refinement (7.36, 7.32, 7.27 W at 250, 400, 800 faces).
- 549 3. As a first-shield coolant at 9.4 W, nitrogen gives 5 times helium's heat-  
550 transfer coefficient in laminar flow and 6 in turbulent flow and holds the  
551 plate within about 4 K of its inlet at  $Re = 2300$  against helium's 21 K;  
552 helium at 1 bar lets the plate mean float above 77 K below  $Re \approx 970$   
553 and its hottest face above 77 K in every constant-property laminar run  
554 (76.9 K at  $Re = 2300$  with variable properties) and needs 38 times the  
555 pumping power, but at 18 bar its pumping power falls below nitrogen's  
556 by 8.5 times with heat transfer unchanged. At each fluid's optimum  
557 nitrogen needs  $1.0 \mu\text{W}$  at  $Re = 500$ ; helium at 18 bar needs  $0.6 \mu\text{W}$  on  
558 the plate-mean criterion and 4 to  $31 \mu\text{W}$  on the hottest-face criterion.  
559 The ranking depends on the basis, the operating point and the wall  
560 criterion.
- 561 4. The ideal ledger favours nitrogen by 21 % for a bare shield and helium  
562 by 9 % with MLI; with hardware efficiencies and the liquefaction en-  
563 ergy of the nitrogen consumed, the equivalent room-temperature power  
564 favours helium by 36–51 % at the nominal shield temperatures and by  
565 13–56 % at the coupled operating points, an advantage that exists only  
566 where the helium plate is actually held near 50 K. Nitrogen becomes

567 competitive when the  $77 \rightarrow 4$  K harness conduction is cut by about a  
568 factor of three, the lever of Eq. (8).  
569 5. The comparison rests on code-verified numbers: NIST properties at a  
570 stated reference state, a forced-convection basis with gravity-on sensi-  
571 tivity, a mass-flux-weighted mixing-cup temperature closing the energy  
572 balance to 0.2 %, wall temperatures read from the wall itself, and row-  
573 normalised view factors reproducing the closed forms to 4 %. That  
574 the shield choice and the harness design are not separable follows from  
575 those numbers; experimental validation of the cold-plate model is the  
576 next step.

#### 577 **CRedit authorship contribution statement**

578 **Imerson Joao:** Conceptualization, Methodology, Software, Investiga-  
579 tion, Formal analysis, Visualization, Writing – original draft, Writing – re-  
580 view & editing.

#### 581 **Declaration of competing interest**

582 The author declares no known competing financial interests or personal  
583 relationships that could have influenced the work reported in this paper.

#### 584 **Data availability**

585 All OpenFOAM cases (enclosure baseline and variants, cold-plate sweep,  
586 grid studies), the post-processing and figure scripts, and the tabulated results  
587 are available at  
588 [github.com/Imerson/cryostat-first-shield-He-vs-LN2](https://github.com/Imerson/cryostat-first-shield-He-vs-LN2); an archived copy will be de-  
589 posited on Zenodo on acceptance.

#### 590 **Acknowledgements**

591 The author thanks Dr. Konstantina Vogiatzaki and Dr. Natalia Ares for  
592 supervision and discussions.

## 593 Declaration of generative AI in scientific writing

594 During the preparation of this work the author used Claude (Anthropic)  
595 to assist with editing, revision and consistency checking of the manuscript  
596 and with scripting the simulation campaign. The author reviewed and edited  
597 all content and takes full responsibility for the published article.

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